Projections, symmetric matrices and quadratic forms

Linear Algebra · Lecture 8

Last time (Lecture 7): the eigenvalues of $A \in \mathbb{R}^{n\times n}$ are the n complex roots of p_A (Lecture 7, Example 6: $1,\ 2\pm i$); A is diagonalizable $\iff$ $\mathbb{R}^n$ has a basis of eigenvectors of A. Earlier results are cited where they are used. $I = I_n$, $e_j = \mathrm{col}_j(I)$.

1. Orthogonal projections

Definition 1. Let W be a subspace of $\mathbb{R}^n$ and $y \in \mathbb{R}^n$. A vector $\hat{w} \in W$ with $y - \hat{w} \perp W$ (Lecture 6, Definition 4: $(y-\hat{w})'w = 0$ for all $w \in W$) is an **orthogonal projection** of y onto W. There is exactly one, **the** orthogonal projection: $y = \hat{w} + (y - \hat{w})$ with $\hat{w} \in W$, $y - \hat{w} \in W^\perp$ for exactly one $\hat{w}$ (Lecture 6, Theorem 5(c): $y = w + v$, $w \in W$, $v \in W^\perp$, in exactly one way).

Theorem 2. Let W be a subspace of $\mathbb{R}^n$, $y \in \mathbb{R}^n$, and $\hat{w}$ the orthogonal projection of y onto W. Then $\|y - w\|^2 = \|y - \hat{w}\|^2 + \|\hat{w} - w\|^2$ $(w \in W)$. So $\|y - w\| > \|y - \hat{w}\|$ for $w \in W$, $w \neq \hat{w}$: $\hat{w}$ is the vector of W closest to y, and $\|y - \hat{w}\|$ is the **distance** from y to W.

Proof.

For $w \in W$: $\hat{w} - w \in W$ and $y - \hat{w} \perp W$, so $y - \hat{w} \perp \hat{w} - w$:

$$\|y - w\|^2 = \|(y - \hat{w}) + (\hat{w} - w)\|^2 = \|y - \hat{w}\|^2 + \|\hat{w} - w\|^2 \quad \text{(Lecture 6, Theorem 2: Pythagoras)}$$
$$w \neq \hat{w} \implies \|\hat{w} - w\|^2 > 0 \implies \|y - w\|^2 > \|y - \hat{w}\|^2 \quad \text{(Lecture 6, Definition 1)}$$

□

Theorem 3. Let $A \in \mathbb{R}^{n\times p}$ with $\operatorname{rank} A = p$. Then $A'A$ is invertible. Let $P = A(A'A)^{-1}A' \in \mathbb{R}^{n\times n}$.

(a) For every $y \in \mathbb{R}^n$, Py is the orthogonal projection of y onto $\mathcal{C}(A)$.

(b) $P' = P = P^2$ and $(I-P)' = I - P = (I-P)^2$: P and $I - P$ are symmetric and **idempotent**.

(c) $PA = A$ and $(I - P)A = 0$.

Proof.

$A'A$ is invertible ($\operatorname{rank}(A'A) = \operatorname{rank} A$; rank p $\iff$ invertible):

$$\operatorname{rank}(A'A) = \operatorname{rank} A = p,\ A'A \in \mathbb{R}^{p\times p} \quad \text{(Lecture 4, Theorems 16(b), 12(d))}$$

(a) $Py \in \mathcal{C}(A)$, $y - Py \perp \mathcal{C}(A)$ (Lecture 6, Theorem 5(a): $y \perp \mathcal{C}(A) \iff A'y = 0$):

$$Py = A\Big((A'A)^{-1}A'y\Big) \in \mathcal{C}(A) \quad \text{(Lecture 4, Definition 7: } \mathcal{C}(A) = \{Ax : x \in \mathbb{R}^p\}\text{)}$$
$$A'(y - Py) = A'y - A'A(A'A)^{-1}A'y = 0 \quad \text{(Lecture 1, Theorem 3(a))}$$

(b) (transposes: Lecture 1, Theorem 8):

$$\Big((A'A)^{-1}\Big)' = \Big((A'A)'\Big)^{-1} = (A'A)^{-1} \quad \text{(Lecture 2, Theorem 11(c))}$$
$$P' = A\Big((A'A)^{-1}\Big)'A' = P, \qquad (I-P)' = I - P' = I - P \quad \text{(Lecture 1, Theorem 8)}$$
$$P^2 = A(A'A)^{-1}(A'A)(A'A)^{-1}A' = P \quad \text{(Lecture 1, Theorem 3(a))}$$
$$(I-P)^2 = I - 2P + P^2 = I - P \quad \text{(Lecture 1, Theorem 3(a))}$$

(c):

$$PA = A(A'A)^{-1}A'A = A, \qquad (I-P)A = A - PA = 0 \quad \text{(Lecture 1, Theorem 3(a))}$$

□

Theorem 4. Let W be a subspace of $\mathbb{R}^n$.

(a) There is exactly one $P_W \in \mathbb{R}^{n\times n}$ such that $P_W y$ is the orthogonal projection of y onto W for every $y \in \mathbb{R}^n$: the **projection matrix** for W. $P_{\{0\}} = 0$. If $W \neq \{0\}$: $P_W = A(A'A)^{-1}A'$ for every $A \in \mathbb{R}^{n\times p}$ with $\operatorname{rank} A = p$ and $\mathcal{C}(A) = W$, and $P_W = QQ'$ if the columns of Q form an orthonormal basis of W.

(b) $P_W' = P_W = P_W^2$; $\mathcal{C}(P_W) = W$; $\operatorname{rank} P_W = \operatorname{tr} P_W = \dim W$; $\|P_W y\|^2 = y'P_W y$ $(y \in \mathbb{R}^n)$; $P_{W^\perp} = I - P_W$.

(c) If $P \in \mathbb{R}^{n\times n}$ and $P' = P = P^2$, then $P = P_{\mathcal{C}(P)}$.

Proof. (a)

- At most one: if $Py = \tilde{P}y$ (the projection, Definition 1) for all y, then $P = \tilde{P}$ (Lecture 1, Proposition 11(b)). $W = \{0\}$: $y - 0 \perp \{0\}$, so $0y = 0$ is the projection.
- $W \neq \{0\}$: a basis $a_1, \ldots, a_p$ of W (Lecture 4, Theorem 4) gives $A = [a_1 \ \cdots \ a_p]$, $\operatorname{rank} A = p$ (Lecture 4, Theorem 12(b): independent columns), $\mathcal{C}(A) = W$ (Lecture 4, Definition 7). For every such A, $A(A'A)^{-1}A'y$ is the projection of y onto W (Theorem 3(a)): P_W exists and equals $A(A'A)^{-1}A'$.
- The orthonormal (so nonzero, orthogonal) columns of $Q \in \mathbb{R}^{n\times p}$ are independent (Lecture 6, Theorem 7(b)): $\operatorname{rank} Q = p$ (Lecture 4, Theorem 12(b)), $\mathcal{C}(Q) = W$, $Q'Q = I_p$ (Lecture 6, Definition 8); so $P_W = Q(Q'Q)^{-1}Q' = QQ'$.

(b) $W = \{0\}$: $P_W = 0$, $\mathcal{C}(0) = \{0\}$, $\operatorname{rank} 0 = \operatorname{tr} 0 = 0$. $W \neq \{0\}$: A as in (a), $p = \dim W$:

$$
\begin{aligned}
P_W' = P_W = P_W^2 &\qquad \text{(Theorem 3(b))}\\
\mathcal{C}(P_W) = \mathcal{C}\Big(A(A'A)^{-1}A'\Big) \subseteq \mathcal{C}(A) = W &\qquad \text{(Lecture 4, Theorem 15(a): } \mathcal{C}(FG) \subseteq \mathcal{C}(F))\\
W = \mathcal{C}(P_W A) \subseteq \mathcal{C}(P_W) &\qquad \text{(Theorem 3(c); Lecture 4, Theorem 15(a))}\\
\operatorname{rank} P_W = \dim \mathcal{C}(P_W) = p &\qquad \text{(Lecture 4, Definition 10: } \operatorname{rank} F = \dim \mathcal{C}(F))\\
p = \operatorname{tr} I_p = \operatorname{tr}\Big((A'A)^{-1}A'A\Big) = \operatorname{tr} P_W &\qquad \text{(Lecture 1, Definition 9, Theorem 10(b))}\\
\|P_W y\|^2 = (P_W y)'P_W y = y'P_W'P_W y = y'P_W y &\qquad \text{(Lecture 6, Definition 1; Lecture 1, Theorem 8)}
\end{aligned}
$$

(b) $P_{W^\perp}$, every W ($W^\perp$ is a subspace, Lecture 6, Theorem 5(a): W is spanned by a basis, or by 0):

$$
\begin{aligned}
&(I - P_W)y = y - P_W y \in W^\perp, \qquad y - (I - P_W)y = P_W y \in W \qquad \text{(Definition 1)}\\
&w \in W,\ v \in W^\perp \implies w'v = v'w = 0, \text{ so } P_W y \perp W^\perp \qquad \text{(Lecture 1, Theorem 8)}
\end{aligned}
$$

So $(I - P_W)y$ is the projection of y onto $W^\perp$ (Definition 1): $P_{W^\perp} = I - P_W$ by (a).

(c) For $v \in \mathbb{R}^n$ and every vector Pz of $\mathcal{C}(P)$ (Lecture 4, Definition 7); $P' = P = P^2$:

$$
Pv \in \mathcal{C}(P), \qquad (Pz)'(v - Pv) = z'P'(v - Pv) = z'(P - P^2)v = 0 \qquad \text{(Lecture 1, Theorem 8)}
$$

So Pv is the projection of v onto $\mathcal{C}(P)$ (Definition 1): $P = P_{\mathcal{C}(P)}$ by (a).

□

Example 5. Let $\mathbf{1} = \mathbf{1}_n = (1, \ldots, 1)'$, $\bar{J}_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$, $C_n = I_n - \bar{J}_n$ (Lecture 1, Definition 14) and $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$. Find $P_{\operatorname{span}\{\mathbf{1}\}}$, P_{W_0} and $\operatorname{rank} P_{W_0}$, and the constant vector $c\mathbf{1}$ closest to $y \in \mathbb{R}^n$.

Solution. $A = \mathbf{1}_n \neq 0$ is independent (Lecture 3, Proposition 12(a)), so $\operatorname{rank} A = 1$ (Lecture 4, Theorem 12(b)); $A'A = \mathbf{1}'\mathbf{1} = n$; so $P_{\operatorname{span}\{\mathbf{1}\}} = \mathbf{1}\, n^{-1}\mathbf{1}' = \bar{J}_n$ (Theorem 4(a)). By Lecture 6, Theorem 5(a) ($y \perp \operatorname{span}\{\mathbf{1}\} \iff \mathbf{1}'y = 0$), $W_0 = \operatorname{span}\{\mathbf{1}\}^\perp$; so $P_{W_0} = I_n - \bar{J}_n = C_n$, and $\operatorname{rank} C_n = \operatorname{tr} C_n = n - n \cdot \frac{1}{n} = n - 1$ (Theorem 4(b); Lecture 1, Definition 9, Theorem 10(a)). The projection of y onto the constant vectors $\operatorname{span}\{\mathbf{1}\}$ is $\bar{J}_n y = \bar{y}\mathbf{1}$ ($\bar{y} = \frac{1}{n}\mathbf{1}'y$, Lecture 1, Example 16); so

(Theorem 2) $\sum_i (y_i - c)^2 = \|y - c\mathbf{1}\|^2 > \|y - \bar{y}\mathbf{1}\|^2 = \sum_i (y_i - \bar{y})^2$ for $c \neq \bar{y}$. For $y = (9,3,5,3)'$: $\bar{y} = 5$, $C_4 y = (4,-2,0,-2)'$, $y = 5\,\mathbf{1}_4 + (4,-2,0,-2)'$ with $\mathbf{1}_4'(4,-2,0,-2)' = 0$.

Example 6. Let $a_1 = \mathbf{1}_4$, $a_2 = (2,0,2,0)'$, $a_3 = (3,3,1,1)'$, $A = [a_1\ a_2\ a_3] \in \mathbb{R}^{4\times 3}$, $W = \mathcal{C}(A)$ and $y = (9,3,5,3)'$. Find $(A'A)^{-1}$ by cofactors, $P = A(A'A)^{-1}A'$, the vector of W closest to y, and the distance from y to W.

Solution.

$A'A$ (Lecture 1, Theorem 4(d)), its cofactors, $(A'A)^{-1} = \frac{1}{\det(A'A)} \operatorname{adj}(A'A)$ (Lecture 5, Theorem 14):

$$A'A = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}, \qquad \begin{array}{lll} c_{11} = \det\left(\begin{smallmatrix} 8 & 8 \\ 8 & 20 \end{smallmatrix}\right) = 96, & c_{12} = -\det\left(\begin{smallmatrix} 4 & 8 \\ 8 & 20 \end{smallmatrix}\right) = -16, & c_{13} = \det\left(\begin{smallmatrix} 4 & 8 \\ 8 & 8 \end{smallmatrix}\right) = -32 \\ c_{22} = \det\left(\begin{smallmatrix} 4 & 8 \\ 8 & 20 \end{smallmatrix}\right) = 16, & c_{23} = -\det\left(\begin{smallmatrix} 4 & 4 \\ 8 & 8 \end{smallmatrix}\right) = 0, & c_{33} = \det\left(\begin{smallmatrix} 4 & 4 \\ 4 & 8 \end{smallmatrix}\right) = 16 \end{array}$$

$c_{ji} = c_{ij}$ (symmetric); along row 1 (Lecture 5, Theorem 13(a)), $\det(A'A) = 4\cdot 96 + 4\cdot(-16) + 8\cdot(-32) = 64$:

$$(A'A)^{-1} = \frac{1}{64}\begin{pmatrix} 96 & -16 & -32 \\ -16 & 16 & 0 \\ -32 & 0 & 16 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

So $\operatorname{rank} A = \operatorname{rank}(A'A) = 3$ (Lecture 4, Theorems 16(b), 12(d)): Theorem 3 applies.

$(A'A)^{-1}A'$ and P:

$$(A'A)^{-1}A' = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 2 & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}$$

$$P = A(A'A)^{-1}A' = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}\frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix}$$

Py, $(I - P)y$, the distance (Theorems 3(a) and 2), a check:

$$Py = \frac{1}{4}\begin{pmatrix} 27+3+5-3 \\ 9+9-5+3 \\ 9-3+15+3 \\ -9+3+5+9 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 6 \\ 2 \end{pmatrix}, \qquad (I-P)y = y - Py = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

The vector of W closest to y is $Py = (8,4,6,2)'$; the distance is $\|y - Py\| = \sqrt{1+1+1+1} = 2$. Check: $A'(I-P)y = (1-1-1+1,\ 2-2,\ 3-3-1+1)' = 0$.

2. Symmetric matrices and the spectral theorem

Theorem 7. Let $A \in \mathbb{R}^{n\times n}$ be symmetric ($A' = A$, Lecture 1, Definition 7). Then each eigenvalue λ of A (root of p_A, Lecture 7, Definition 5) is real, and $Ax = \lambda x$ for some $x \in \mathbb{R}^n$, $x \neq 0$.

Proof. Let $\mu = \alpha + i\beta$ ($\alpha, \beta \in \mathbb{R}$) be a root of p_A, and $Y = (A - \alpha I)^2 \in \mathbb{R}^{n\times n}$; $p_A(c) = \det(cI - A)$ for $c \in \mathbb{R}$ (Lecture 7, Section 2); $\det(FG) = \det F \det G$ (Lecture 5, Theorem 12(b)).

For $s \in \mathbb{R}$ and $x \in \mathbb{R}^n$; $\|x\|^2 = x'x$ (Lecture 6, Definition 1):

$$\begin{aligned} \Big((\alpha+s)I - A\Big)\Big((\alpha-s)I - A\Big) &= (\alpha I - A)^2 - s^2 I = Y - s^2 I && \text{(Lecture 1, Theorem 3(a))} \\ p_A(\alpha+s)\, p_A(\alpha-s) &= \det(Y - s^2 I) && \text{(Lecture 5, Theorem 12(b))} \\ x'(Y + \beta^2 I)x &= \Big((A-\alpha I)x\Big)'(A-\alpha I)x + \beta^2 x'x && (A' = A\text{; Lecture 1, Theorem 8}) \\ &= \|(A-\alpha I)x\|^2 + \beta^2\|x\|^2 && \text{(Lecture 6, Definition 1)} \end{aligned}$$

- As $\det(Y - s^2 I) = (-1)^n \det(s^2 I - Y)$ (Lecture 5, Theorem 12(c): $\det(cF) = c^n \det F$; Lecture 7, Section 2), the second identity says that the polynomials $p_A(\alpha + t)\, p_A(\alpha - t)$ (expand the powers of $\alpha \pm t$) and $(-1)^n p_Y(t^2)$ agree at every $t = s \in \mathbb{R}$. So they are equal (Lecture 7, F2: equal values at infinitely many $z \Rightarrow$ equal), and at $t = i\beta$ (Lecture 7, F4: $(fg)(z) = f(z)g(z)$): $p_A(\mu)\, p_A(\alpha - i\beta) = (-1)^n p_Y(-\beta^2) = \det(Y + \beta^2 I)$; and $p_A(\mu) = 0$.
- If $\beta \neq 0$, the third identity gives $x'(Y + \beta^2 I)x > 0$, so $(Y + \beta^2 I)x \neq 0$, for every $x \neq 0$: $\det(Y + \beta^2 I) \neq 0$ (Lecture 2, Theorem 15; Lecture 5, Theorem 12(a): invertible $\iff \det \neq 0$) ✘ Hence $\beta = 0$.
- $\lambda = \mu = \alpha \in \mathbb{R}$ and $\det(\lambda I - A) = p_A(\lambda) = 0$: $Ax = \lambda x$ for some $x \neq 0$ in $\mathbb{R}^n$ (Lecture 7, Theorem 2: for $\lambda \in \mathbb{R}$, $\det(\lambda I - A) = 0 \iff Ax = \lambda x$ for some $x \neq 0$). □

Lecture 7, Example 6 (eigenvalues $1, 2 \pm i$) is not symmetric. Most matrices of statistics here whose eigenvalues we use are symmetric, so their eigenvalues are real (Theorem 7): $A'A$, projection matrices ($\bar{J}_n$, C_n), covariance and correlation matrices. The others, $A^{-1}B$ (A, B symmetric, $x'Ax > 0$ for $x \neq 0$), need not be symmetric but have real eigenvalues too (end of Section 3).

Theorem 8. Let $A \in \mathbb{R}^{n \times n}$ be symmetric, $Ax = \lambda x$ and $Ay = \mu y$ ($x, y \in \mathbb{R}^n$), $\lambda \neq \mu$. Then $x'y = 0$.

Proof.

One line (Lecture 1, Theorem 8; $\lambda \neq \mu$):

$$\lambda x'y = (Ax)'y = x'A'y = x'Ay = \mu x'y \implies (\lambda - \mu)x'y = 0 \implies x'y = 0$$

□

Theorem 9 (Spectral theorem)**.** Let $A \in \mathbb{R}^{n \times n}$ be symmetric.

(a) $\mathbb{R}^n$ has an orthonormal basis $q_1, \ldots, q_n$ of eigenvectors of A.

(b) If $q_1, \ldots, q_n \in \mathbb{R}^n$ are orthonormal, $Aq_i = \lambda_i q_i$, $Q = [q_1 \ \cdots \ q_n]$ and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$: Q is orthogonal and $Q'AQ = \Lambda$, $A = Q\Lambda Q'$; $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A; $\operatorname{rank} A = \#\{i : \lambda_i \neq 0\}$; and $\dim \mathcal{N}(A - cI) = a_A(c)$ for $c \in \mathbb{R}$.

(c) Orthonormal bases of the eigenspaces of A, put side by side in any order, are such $q_1, \ldots, q_n$.

(d) For such Q, Λ, with the distinct eigenvalues $c_1, \ldots, c_k$, $W_j = \mathcal{N}(A - c_j I)$ and $E_j = P_{W_j}$: $E_j = \sum_{i:\lambda_i = c_j} q_i q_i'$ and $A = \lambda_1 q_1 q_1' + \cdots + \lambda_n q_n q_n' = c_1 E_1 + \cdots + c_k E_k$.

Proof. (a) By induction on n: $Q'AQ$ is diagonal for an orthogonal Q. $n = 1$: $Q = (1)$. Let $n \geqslant 2$.

(a), step 1: $p_A = (t - \lambda_1) \cdots (t - \lambda_n)$ (Lecture 7, Definition 5) has a root λ, real, with $Ax = \lambda x$, $x \neq 0$ (Theorem 7); $u = x/\|x\|$; $[u\ V]$ orthogonal (Lecture 6, Corollary 11(c), $p = 1$); blocks multiply and transpose block by block (Lecture 5, Theorem 2(b),(c)):

$$\begin{aligned}
Au = \lambda u, \quad u'u = 1 & \\
[u\ V]'[u\ V] = I \implies V'u = 0,\ V'V = I_{n-1} & \qquad \text{(Lecture 6, Definition 8)} \\
V'Au = \lambda V'u = 0, \quad u'Au = \lambda & \\
u'AV = (V'A'u)' = (V'Au)' = 0 & \qquad (A' = A) \\
[u\ V]'A[u\ V] = \begin{pmatrix} u'Au & u'AV \\ V'Au & V'AV \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & B \end{pmatrix} & \qquad (B = V'AV) \\
B' = V'A'V = B & \qquad (A' = A;\ \text{Lecture 1, Theorem 8})
\end{aligned}$$

(a), step 2: induction, $R'BR = \Lambda_2$ diagonal with R orthogonal; $S = \left(\begin{smallmatrix} 1 & 0 \\ 0 & R \end{smallmatrix}\right)$, $Q = [u\ V]S$:

$$\begin{aligned}
Q'Q = S'[u\ V]'[u\ V]S = S'S = \begin{pmatrix} 1 & 0 \\ 0 & R'R \end{pmatrix} = I & \qquad \text{(Lecture 1, Theorem 8; blocks)} \\
Q'AQ = S' \begin{pmatrix} \lambda & 0 \\ 0 & B \end{pmatrix} S = \begin{pmatrix} \lambda & 0 \\ 0 & R'BR \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \Lambda_2 \end{pmatrix} & \qquad \text{(blocks)}
\end{aligned}$$

With $\Lambda = Q'AQ$ and the columns q_i of Q: $AQ = QQ'AQ = Q\Lambda$ (Lecture 6, Theorem 9(b): $QQ' = I$), so $Aq_i = \lambda_i q_i$ (Lecture 1, Theorem 4(a), Proposition 13(a)); $Q'Q = I$: $q_1, \dots, q_n$ are orthonormal (Lecture 6, Definition 8), independent (Lecture 6, Theorem 7(b)), a basis of $\mathbb{R}^n$ (Lecture 4, Theorem 5(d): n independent vectors of $\mathbb{R}^n$).

(b) uses $\operatorname{rank}(QFQ') = \operatorname{rank} F$ for invertible Q, Q' (Lecture 4, Theorem 15(d)), $\operatorname{rank} F + \dim \mathcal{N}(F) = n$ for $F \in \mathbb{R}^{n\times n}$ (Lecture 4, Theorem 11), and $Q^{-1}AQ = \Lambda \iff Aq_j = \lambda_j q_j$ (all j), the λ_j then being the eigenvalues (Lecture 7, Theorem 14(a)).

(b) $(Q'Q)_{ij} = q_i' q_j$ (Lecture 1, Theorem 4(d)), $Aq_j = \lambda_j q_j$; then $A - cI = Q(\Lambda - cI)Q'$ ($c \in \mathbb{R}$):

$$
\begin{aligned}
Q'Q = I, \quad Q^{-1} = Q', \quad QQ' = I & \qquad \text{(Lecture 6, Definition 8, Theorem 9(b))}\\
Q'AQ = Q^{-1}AQ = \Lambda, \quad A = Q\Lambda Q' & \qquad \text{(Lecture 7, Theorem 14(a))}\\
\operatorname{rank}(A - cI) = \operatorname{rank}(\Lambda - cI) = \#\{i : \lambda_i \neq c\} & \qquad \text{(Lecture 4, Theorem 15(d); below)}\\
\dim \mathcal{N}(A - cI) = n - \operatorname{rank}(A - cI) = \#\{i : \lambda_i = c\} & \qquad \text{(Lecture 4, Theorem 11)}\\
= a_A(c) & \qquad \text{(Lecture 7, Definition 5)}
\end{aligned}
$$

$\lambda_1, \dots, \lambda_n$ are the eigenvalues (Lecture 7, Theorem 14(a)). The nonzero columns $(\lambda_i - c)e_i$ of $\Lambda - cI$ are orthogonal, so independent (Lecture 6, Theorem 7(b)), and span $\mathcal{C}(\Lambda - cI)$ (Lecture 4, Definition 7). For $c = 0$: $\operatorname{rank} A = \#\{i : \lambda_i \neq 0\}$.

(c) The distinct eigenvalues $c_1, \dots, c_k$ are real (Theorem 7), and $\dim \mathcal{N}(A - c_j I) = a_A(c_j) \geqslant 1$ by (a), (b); orthonormal bases of the $\mathcal{N}(A - c_j I)$, of $a_A(c_j)$ vectors (Lecture 6, Corollary 11(a): a subspace has an orthonormal basis; Lecture 4, Theorem 4: every basis has dim vectors), orthogonal to each other (Theorem 8), give $a_A(c_1) + \dots + a_A(c_k) = n$ orthonormal eigenvectors (Lecture 7, Definition 5: the n roots of p_A, with multiplicity).

(d) Q_j: the columns q_i with $\lambda_i = c_j$; outer products (Lecture 1, Theorem 4(d)):

$$
\begin{aligned}
Q_j'Q_j = I,\ \mathcal{C}(Q_j) \subseteq W_j,\ a_A(c_j) = \dim W_j \text{ columns} & \qquad \text{(Lecture 6, Definition 8; (b))}\\
\text{the columns of } Q_j \text{ are independent} & \qquad \text{(Lecture 6, Theorem 7(b))}\\
\text{so they form a basis of } W_j & \qquad \text{(Lecture 4, Theorem 5(d))}\\
E_j = Q_j Q_j' = \textstyle\sum_{i:\lambda_i = c_j} q_i q_i' & \qquad \text{(Theorem 4(a))}\\
A = (Q\Lambda)Q' = \lambda_1 q_1 q_1' + \dots + \lambda_n q_n q_n' & \qquad \text{(Lecture 1, Proposition 13(a))}
\end{aligned}
$$

Grouping the terms by the value c_j of λ_i: $A = c_1 E_1 + \dots + c_k E_k$.

□

Example 10. Find an orthogonal Q and a diagonal Λ with $Q'AQ = \Lambda$, and the matrices E_j of Theorem 9(d), for

$$
A = \begin{pmatrix} 11 & 2 & 6 & 6 \\ 2 & 11 & 6 & 6 \\ 6 & 6 & 11 & 2 \\ 6 & 6 & 2 & 11 \end{pmatrix}
$$

Solution.

p_A: $R_1 \leftarrow R_1 - R_2$, $R_3 \leftarrow R_3 - R_4$; take $t - 9$ out of rows 1, 3 (Lecture 5, Theorem 8(a),(b)):

$$
\det(tI - A) = \det \begin{pmatrix} t-11 & -2 & -6 & -6 \\ -2 & t-11 & -6 & -6 \\ -6 & -6 & t-11 & -2 \\ -6 & -6 & -2 & t-11 \end{pmatrix} = (t-9)^2 \det \begin{pmatrix} 1 & -1 & 0 & 0 \\ -2 & t-11 & -6 & -6 \\ 0 & 0 & 1 & -1 \\ -6 & -6 & -2 & t-11 \end{pmatrix}
$$

p_A, continued: $\mathrm{col}_2 \leftarrow \mathrm{col}_2 + \mathrm{col}_1$, $\mathrm{col}_4 \leftarrow \mathrm{col}_4 + \mathrm{col}_3$; expand along rows 1, 2 (Lecture 5, Theorems 8(b), 13):

$$\det(tI-A) = (t-9)^2 \det\begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & t-13 & -6 & -12 \\ 0 & 0 & 1 & 0 \\ -6 & -12 & -2 & t-13 \end{pmatrix} = (t-9)^2 \det\begin{pmatrix} t-13 & -6 & -12 \\ 0 & 1 & 0 \\ -12 & -2 & t-13 \end{pmatrix}$$
$$= (t-9)^2\Big((t-13)^2 - 144\Big) = (t-9)^2(t-25)(t-1)$$

So $p_A = (t-25)(t-9)^2(t-1)$ (Lecture 7, Definition 5: the eigenvalues are its roots): $c_1 = 25$, $c_2 = 9$ (twice), $c_3 = 1$.

$\lambda = 25$, 1: $\dim \mathcal{N}(A-25I) = \dim \mathcal{N}(A-I) = 1$ (Theorem 9(b)); row sums 25:

$$A\mathbf{1}_4 = 25\,\mathbf{1}_4, \qquad v = (1,1,-1,-1)'$$
$$Av = (11+2-12,\ 2+11-12,\ 12-11-2,\ 12-2-11)' = v$$

So $q_1 = \frac{1}{2}\mathbf{1}_4$ and $q_4 = \frac{1}{2}v$ are orthonormal bases of $\mathcal{N}(A-25I)$ and $\mathcal{N}(A-I)$ (Lecture 4, Theorem 5(d): one nonzero vector of a line is a basis).

$\lambda = 9$: reduce $A-9I$; special solutions, a basis of $\mathcal{N}(A-9I)$ (Lecture 4, Theorem 8(c)):

$$A-9I = \begin{pmatrix} 2&2&6&6 \\ 2&2&6&6 \\ 6&6&2&2 \\ 6&6&2&2 \end{pmatrix} \xrightarrow[R_4\leftarrow R_4-R_3]{R_2\leftarrow R_2-R_1} \begin{pmatrix} 2&2&6&6 \\ 0&0&0&0 \\ 6&6&2&2 \\ 0&0&0&0 \end{pmatrix} \xrightarrow{R_3\leftarrow R_3-3R_1} \begin{pmatrix} 2&2&6&6 \\ 0&0&0&0 \\ 0&0&-16&-16 \\ 0&0&0&0 \end{pmatrix}$$
$$\xrightarrow{R_2\leftrightarrow R_3} \begin{pmatrix} 2&2&6&6 \\ 0&0&-16&-16 \\ 0&0&0&0 \\ 0&0&0&0 \end{pmatrix} \xrightarrow[R_2\leftarrow -\frac{1}{16}R_2]{R_1\leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1&1&3&3 \\ 0&0&1&1 \\ 0&0&0&0 \\ 0&0&0&0 \end{pmatrix} \xrightarrow{R_1\leftarrow R_1-3R_2} \begin{pmatrix} 1&1&0&0 \\ 0&0&1&1 \\ 0&0&0&0 \\ 0&0&0&0 \end{pmatrix}$$

Free unknowns x_2, x_4; special solutions $u_2 = (-1,1,0,0)'$, $u_3 = (0,0,-1,1)'$, orthogonal: so $q_2 = \frac{1}{\sqrt{2}}u_2$, $q_3 = \frac{1}{\sqrt{2}}u_3$ form an orthonormal basis of $\mathcal{N}(A-9I)$.

$q_1, q_4 \perp q_2, q_3$ (Theorem 8), $q_1'q_4 = 0$: Q, Λ (Theorem 9(b)); E_1, E_2, E_3 (Theorem 9(d)):

$$Q = \begin{pmatrix} 1/2 & -1/\sqrt{2} & 0 & 1/2 \\ 1/2 & 1/\sqrt{2} & 0 & 1/2 \\ 1/2 & 0 & -1/\sqrt{2} & -1/2 \\ 1/2 & 0 & 1/\sqrt{2} & -1/2 \end{pmatrix}, \qquad \Lambda = \mathrm{diag}(25,9,9,1)$$

$$E_1 = q_1q_1' = \tfrac{1}{4}\mathbf{1}_4\mathbf{1}_4' = \bar{J}_4, \qquad E_2 = q_2q_2' + q_3q_3' = \frac{1}{2}\begin{pmatrix} 1&-1&0&0 \\ -1&1&0&0 \\ 0&0&1&-1 \\ 0&0&-1&1 \end{pmatrix}, \qquad E_3 = q_4q_4' = \tfrac{1}{4}vv'$$

$E_1 + E_2 + E_3 = QQ' = I$, $A = 25E_1 + 9E_2 + E_3$; entries $(1,1)$, $(1,2)$, $(1,3)$: $\frac{25}{4} + \frac{9}{2} + \frac{1}{4} = 11$, $\frac{25}{4} - \frac{9}{2} + \frac{1}{4} = 2$, $\frac{25}{4} - \frac{1}{4} = 6$.

Theorem 11. Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then $A^2 = A \iff$ every eigenvalue of A is 0 or 1. In that case $a_A(1) = \operatorname{rank} A = \operatorname{tr} A$.

Proof.

An eigenvalue λ has $Ax = \lambda x$, $x \in \mathbb{R}^n$, $x \neq 0$ (Theorem 7); $A = Q\Lambda Q'$ as in Theorem 9(a),(b); $cx = 0$, $x \neq 0 \Rightarrow c = 0$ (Lecture 3, Proposition 2):

$$\begin{aligned}
A^2 = A &\implies \lambda x = Ax = A^2x = \lambda Ax = \lambda^2 x && \text{(Lecture 1, Theorem 3(a))}\\
&\implies (\lambda - \lambda^2)x = 0 \implies \lambda \in \{0,1\} && \text{(Lecture 3, Proposition 2)}\\
\lambda_i \in \{0,1\} &\implies \Lambda^2 = \Lambda && \text{(Lecture 1, Proposition 13(b))}\\
&\implies A^2 = Q\Lambda Q'Q\Lambda Q' = Q\Lambda^2 Q' = A && (Q'Q = I\text{, Lecture 6, Definition 8})\\
\operatorname{rank} A = \#\{i : \lambda_i \neq 0\} &= \#\{i : \lambda_i = 1\} = a_A(1) && \text{(Theorem 9(b))}\\
&= \lambda_1 + \cdots + \lambda_n = \operatorname{tr} A && \text{(Lecture 7, Theorem 9)}
\end{aligned}$$

□

So a projection matrix P_W (symmetric and idempotent, Theorem 4(b)) has the eigenvalues 1 ($\dim W$ times) and 0 ($n - \dim W$ times): C_n has 1 ($n-1$ times) and 0 (once) by Example 5, and P of Example 6 has $1, 1, 1, 0$, as $\operatorname{tr} P = \frac{1}{4}(3+3+3+3) = 3$.

3. Quadratic forms

Definition 12. The **quadratic form** of $A \in \mathbb{R}^{n\times n}$ is $q_A(x) = x'Ax = \sum_i \sum_j a_{ij}x_ix_j$ ($x \in \mathbb{R}^n$); that is, $x'Ax = \sum_i a_{ii}x_i^2 + \sum_{i<j}(a_{ij} + a_{ji})x_ix_j$.

As $x'Ax = (x'Ax)' = x'A'x$ (Lecture 1, Definition 7, Theorem 8), $x'Ax = x'Bx$ for the symmetric $B = \frac{1}{2}(A + A')$: quadratic forms are written with symmetric matrices. For symmetric A, a_{ii} is the coefficient of x_i^2, and $a_{ij} = a_{ji}$ is half the coefficient of x_ix_j ($i \neq j$).

Definition 13. A symmetric $A \in \mathbb{R}^{n\times n}$, and q_A, is **positive definite** (PD) if $x'Ax > 0$ for all $x \neq 0$; **positive semidefinite** (PSD) if $x'Ax \geqslant 0$ for all x; **negative definite** (ND) or **negative semidefinite** (NSD) if $-A$ is PD or PSD; **indefinite** if $x'Ax$ takes both positive and negative values. A PD matrix is PSD.

Theorem 14. Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and Q, $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ as in Theorem 9(b) ($Q'AQ = \Lambda$; the λ_i are the eigenvalues). For $x \in \mathbb{R}^n$ and $y = Q'x$: $x'Ax = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$, and $y = 0 \iff x = 0$.

(a) A is PD $\iff$ every $\lambda_i > 0$; A is PSD $\iff$ every $\lambda_i \geqslant 0$.

(b) A is ND $\iff$ every $\lambda_i < 0$; NSD $\iff$ every $\lambda_i \leqslant 0$; indefinite $\iff$ $\lambda_i > 0 > \lambda_j$ for some i, j.

(c) If A is PD, then $\det A > 0$, and A is invertible.

Proof.

The identity ($QQ' = I$, Lecture 6, Theorem 9(b); Lecture 1, Theorems 3(a), 8):

$$x = QQ'x = Qy, \qquad y = Q'x \qquad (\text{so } y = 0 \iff x = 0)$$
$$x'Ax = x'QQ'AQQ'x = (Q'x)'(Q'AQ)(Q'x) = y'\Lambda y = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$$

(a) If every $\lambda_i > 0$ and $x \neq 0$: $y \neq 0$, some $y_i^2 > 0$, so $x'Ax > 0$. Conversely, $x = q_i$ (column i of Q) gives $y = Q'Qe_i = e_i$ (Lecture 6, Definition 8; Lecture 1, Theorem 4(b)) and $q_i'Aq_i = \lambda_i$; so PD $\Rightarrow$ $\lambda_i > 0$. PSD: the same with $\geqslant$.

(b) $(-A)q_i = -\lambda_i q_i$: (a) for $-A$ gives ND and NSD. If $\lambda_i > 0 > \lambda_j$, then $q_i'Aq_i = \lambda_i > 0 > \lambda_j = q_j'Aq_j$: indefinite. If no $\lambda_i < 0$ (no $\lambda_i > 0$), A is PSD (NSD), not indefinite.

(c) $\det A = \lambda_1 \cdots \lambda_n$ (Lecture 7, Theorem 9); invertible $\iff$ $\det A \neq 0$ (Lecture 5, Theorem 12(a)):

$$\det A = \lambda_1 \cdots \lambda_n > 0 \implies A \text{ invertible} \qquad ((a))$$

□

Theorem 15. (a) Let $B \in \mathbb{R}^{n\times n}$ be symmetric and $K \in \mathbb{R}^{n\times k}$. If B is PSD, then $K'BK$ is PSD. If B is PD and $\operatorname{rank} K = k$, then $K'BK$ is PD.

(b) For $A \in \mathbb{R}^{n\times p}$, $A'A$ is PSD, and $A'A$ is PD $\iff$ $\operatorname{rank} A = p$.

Proof.

(a) For $x \in \mathbb{R}^k$; $\operatorname{rank} K = k \iff \mathcal{N}(K) = \{0\}$ (Lecture 4, Theorem 12(b)):

$$\begin{aligned} (K'BK)' &= K'B'K = K'BK && \text{(Lecture 1, Theorem 8)} \\ x'(K'BK)x &= (Kx)'B(Kx) \geqslant 0 && \text{(Lecture 1, Theorem 8; } B \text{ PSD)} \\ \operatorname{rank} K = k,\ x \neq 0 &\implies Kx \neq 0 \implies (Kx)'B(Kx) > 0 && (B \text{ PD}) \end{aligned}$$

(b) I_n is PD ($x'x > 0$ for $x \neq 0$, Lecture 6, Definition 1):

$$\begin{aligned} A'A = A'I_nA &\text{ is PSD, and PD if } \operatorname{rank} A = p && \text{((a) with } I_n, A \text{ for } B, K) \\ A'A \text{ PD} &\implies A'A \text{ invertible} && \text{(Theorem 14(c))} \\ &\implies \operatorname{rank} A = \operatorname{rank}(A'A) = p && \text{(Lecture 4, Theorems 16(b), 12(d))} \end{aligned}$$

□

In Lecture 7, Example 19 ($A \in \mathbb{R}^{4\times 3}$), $A'A$ has the eigenvalues $1, 3, 4$ and AA' the eigenvalues $0, 1, 3, 4$: by Theorem 14(a), $A'A$ is PD, and AA' is PSD but not PD.

Example 16. Classify (a) q_A for A of Example 10; (b) C_n ($n \geqslant 2$); (c) for each $a \in \mathbb{R}$,

$$q(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2 + 2a(x_1x_2 + x_1x_3 + x_2x_4 + x_3x_4)$$

Solution. (a) A of Example 10 has the eigenvalues $25, 9, 9, 1$, all > 0: PD (Theorem 14(a)); with $y = Q'x$, $q_A(x) = 25y_1^2 + 9y_2^2 + 9y_3^2 + y_4^2$.

(b) C_n has the eigenvalues 1 ($n-1$ times) and 0 (after Theorem 11): PSD, not PD (Theorem 14(a)). By Theorem 4(b) ($C_n = P_{W_0}$, Example 5) and $C_ny = y - \bar{y}\mathbf{1}$ (Lecture 1, Example 16), $\sum_i (y_i - \bar{y})^2 = \|C_ny\|^2 = y'C_ny \geqslant 0$, with 0 only for constant $y = \bar{y}\mathbf{1}$.

(c) The matrix of q (Definition 12); four vectors h_i, Ah_i row by row:

$$A = \begin{pmatrix} 1 & a & a & 0 \\ a & 1 & 0 & a \\ a & 0 & 1 & a \\ 0 & a & a & 1 \end{pmatrix}, \qquad h_1 = \mathbf{1}_4, \quad h_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \quad h_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \quad h_4 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$\begin{aligned} Ah_1 &= (1+a+a,\ a+1+a,\ a+1+a,\ a+a+1)' = (1+2a)h_1 \\ Ah_2 &= (1+a-a,\ a+1-a,\ a-1-a,\ a-a-1)' = h_2 \\ Ah_3 &= (1-a+a,\ a-1-a,\ a+1-a,\ -a+a-1)' = h_3 \\ Ah_4 &= (1-a-a,\ a-1+a,\ a-1+a,\ -a-a+1)' = (1-2a)h_4 \end{aligned}$$

$Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$, Λ (Theorem 9(b)), and the classification (Theorem 14):

$$\begin{aligned} h_i'h_j = 0\ (i \neq j),\ h_i'h_i = 4 &\implies \tfrac{1}{2}h_1, \ldots, \tfrac{1}{2}h_4 \text{ orthonormal} && \text{(Lecture 6, Definition 8)} \\ Q'AQ = \Lambda &= \operatorname{diag}(1+2a, 1, 1, 1-2a) && \text{(Theorem 9(b))} \end{aligned}$$

- $|a| < \frac{1}{2}$: all four are > 0: PD.
- $a = \pm\frac{1}{2}$: the eigenvalues are $2, 1, 1, 0$ (in some order): PSD, not PD.
- $|a| > \frac{1}{2}$: $1 + 2a$ and $1 - 2a$ have opposite signs: indefinite.

Theorem 17. Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric, A PD. Then $E'AE = I$ and $E'BE = \Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$ for some invertible $E = [f_1 \ \cdots \ f_n] \in \mathbb{R}^{n\times n}$; and $A^{-1}BE = E\Lambda$: $A^{-1}Bf_i = \lambda_i f_i$, and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $A^{-1}B$, all real.

Proof. • $Q'AQ = D = \text{diag}(d_1, \ldots, d_n)$, Q orthogonal (Theorem 9(a),(b)), every $d_i > 0$ (Theorem 14(a)); $D^{-1/2} = \text{diag}(d_1^{-1/2}, \ldots, d_n^{-1/2})$: $D^{-1/2}Q'AQD^{-1/2} = D^{-1/2}DD^{-1/2} = I$ (Lecture 1, Proposition 13(b)).

- $M = D^{-1/2}Q'BQD^{-1/2}$ is symmetric (Lecture 1, Theorem 8): $R'MR = \Lambda$ is diagonal for an orthogonal R (Theorem 9(a),(b)).
- $E = QD^{-1/2}R$: $E'AE = R'(D^{-1/2}Q'AQD^{-1/2})R = R'R = I$, $E'BE = R'MR = \Lambda$. From $E'(AE) = I$: $AE = (E')^{-1}$, and $E = (E')'$ is invertible (Lecture 2, Theorems 15, 11(c)).
- $BE = (E')^{-1}\Lambda = AE\Lambda$; A is invertible (Theorem 14(c)): $A^{-1}BE = E\Lambda$, $E^{-1}(A^{-1}B)E = \Lambda$. By Lecture 7, Theorem 14(a) with $A^{-1}B$, E for A, P, $A^{-1}Bf_i = \lambda_i f_i$, and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $A^{-1}B$. □

Practice. The Exercises and Problems of the textbook: Chapter 6 on orthogonal projections and projection matrices, Chapter 8 on orthogonal diagonalization and on classifying quadratic forms.